1 **(a)** Class B, because the mean (or median) is higher. **[B1]**

(b) Class B, because the range (or interquartile range) is smaller. **[B1]**

(c) Identifies that the table only gives summary statistics / does not show individual marks **[B1]**

Mentions overlapping ranges / a specific example (e.g. the lowest mark in Class B could be lower than the highest mark in Class A) **[B1]**

[Total Marks: 4]

2 **(a)** Median = 2 **[B1]**

Range = 4 **[B1]**

(b) Team A generally scored more goals because their median is higher ($2 > 1.5$) **[B1]**

Team A was more consistent because their range is smaller ($4 < 5$) **[B1]**

[Total Marks: 4]

3 **(a)** B **[B1]**

(b) 3 **[B1]**

(c) Total students $= 5+10+3+2+7+8+3+2 = 40$ AND Total Grade A $= 5+7 = 12$ **[M1]**

30 **[A1]**

[Total Marks: 4]

4 **(a)** 27 **[B1]**

(b) Upper quartile = 35 OR Lower quartile = 20 **[M1]**

15 **[A1]**

(c) $\frac{7}{15}$ **[B1]**

[Total Marks: 4]

5 **(a)** 27 **[B1]**

(b) 31 **[B1]**

(c) 36 and 20 identified as quartiles **[M1]**

16 **[A1]**

(d) $\frac{6}{15}$ or $\frac{2}{5}$ **[B1]**

[Total Marks: 5]

6 **(a)** Median = 32 **[B1]**

Interquartile range $= 40 - 25 = 15$ **[B1]**

(b) Class A has a higher median score ($32 > 29$), so on average Class A performed better than Class B **[B1]**

Class A has a smaller interquartile range ($15 < 18$), so Class A's scores are more consistent (less spread out) than Class B's **[B1]**

(c) In Class B, 3 students scored above 40 ($41, 46, 49$), and in Class A, 3 students also scored above 40 ($42, 45, 47$, as 40 is not above 40) **[M1]**

No, the claim is incorrect because both classes had the same number (3 students) scoring above 40 **[A1]**

(d) The results are from only one test (or a small sample of 15 students per class) and may not represent overall mathematical ability across all topics **[B1]**

[Total Marks: 7]

7 (a) Identifies the middle value is 7 **[M1]**

Identifies that 3 must appear at least twice as the lowest values: $3, 3$ **[M1]**

Uses the range to find the maximum value: $3 + 11 = 14$ **[M1]**

Uses the mean to find the missing value: $40 - (3 + 3 + 7 + 14) = 13$, yielding $3, 3, 7, 13, 14$ **[A1]**

(b) New sum $= 6 \times 9 = 54$ AND Old sum $= 5 \times 8 = 40$ (or $54 - 40$) **[M1]**

14 **[A1]**

(c) No, because standard deviation measures the spread of all data points around the mean, not just the difference between the extreme values **[B1]**

[Total Marks: 7]

8 (a) $y = 80 - (12 + 28 + 14 + 6) = 20$ **[B1]**

Calls lasting more than 8 minutes in Branch South $= 35 + 20 + 10 = 65$, giving $\dfrac{65}{80 + x} = \dfrac{13}{20}$ **[M1]**

$13(80 + x) = 1300$ giving $80 + x = 100$ **[M1]**

$x = 20$ **[A1]**

(b) Branch North: $\sum fm = 12(2) + 28(6) + 20(10) + 14(14) + 6(18) = 696$ **[M1]**

Branch North estimated mean $= \dfrac{696}{80} = 8.7$ minutes **[A1]**

Branch South: $\sum fm = 15(2) + 20(6) + 35(10) + 20(14) + 10(18) = 960$ **[M1]**

Branch South estimated mean $= \dfrac{960}{100} = 9.6$ minutes **[A1]**

(c) Statement 1 is valid because the shortest possible call duration is just above 0 minutes and the longest possible duration is 20 minutes, giving a maximum possible range of $20 - 0 = 20$ minutes **[B2]**

Statement 2 is not valid because the 35 calls in the interval $8 < t \leq 12$ are grouped, so all

35 calls could lie outside $9 \leq t \leq 11$ (for example, between 8 and 9 minutes) **[B2]**

[Total Marks: 12]

9 **(a)** 40 **[B1]**

(b) 25 (from $50 - 25$) **[B1]**

(c) Group Y was slower on average / had a higher median time ($40 > 35$) **[B1]**

Group Y's times were more consistent / had a smaller IQR ($15 < 25$) or smaller range ($40 < 60$) **[B1]**

(d) Identifies that exactly 25% of Group Y took less than 30 minutes (since lower quartile is 30) **[M1]**

States Group X had a larger proportion, because 30 is between their LQ (25) and median (35), meaning more than 25% took less than 30 minutes **[A1]**

[Total Marks: 6]

10 **(a)** 45 **[B1]**

(b) 10 **[B1]**

(c) Group A students were faster on average because their median time (45 min) is less than Group B (50 min). **[B1]**

Group B times are more consistent / less spread out because their interquartile range (10 min) is smaller than Group A (20 min). **[B1]**

(d) Recognises that 55 is the upper quartile, representing 25% or $\frac{1}{4}$ of the students **[M1]**

20 **[A1]**

[Total Marks: 6]

11 **(a)** 65 **[B1]**

(b) $75 - 40$ **[M1]**

35 **[A1]**

(c) Class X has a higher average score (or median is higher: $65 > 55$) **[B1]**

Class X scores are less spread out / more consistent (or IQR is lower: $30 < 35$, or Range is lower: $60 < 65$) **[B1]**

(d) No, because the ranges overlap (e.g., the maximum score in Class Y is 90, which is higher than the lower quartile of Class X) **[B1]**

[Total Marks: 6]

12 **(a)** $120 < m \leq 140$ **[B1]**

(b) Use of correct midpoints: $110, 130, 150, 170, 190$ **[M1]**

$110 \times 10 + 130 \times 20 + 150 \times 15 + 170 \times 10 + 190 \times 5$ **[M1]**

$= 1100 + 2600 + 2250 + 1700 + 950 = 8600$ **[M1]**

$\frac{8600}{60} = 143.33\ldots$ so 143 (to 3 sf) **[A1]**

(c) The exact masses of the individual mangoes are unknown because the data is grouped **[B1]**

The 15 mangoes could have any combination of masses between 140 g and 160 g, not necessarily exactly at the midpoint **[B1]**

[Total Marks: 7]

13 (a) Sum of goals $= (0 \times 8) + (1 \times 12) + (2 \times x) + (3 \times 6) + (4 \times y) + (5 \times 2)$ OR $12 + 2x + 18 + 4y + 10$ **[M1]**

Equates sum of goals to 40×1.7 OR $\frac{40 + 2x + 4y}{40} = 1.7$ **[M1]**

$40 + 2x + 4y = 68$

2x + 4y = $28(Correctcompletionwithnoerrors)[A1]$

(b) $8 + 12 + x + 6 + y + 2 = 40$ **[M1]**

$x + y = 12$ **[A1]**

Solves simultaneous equations: e.g. $2x + 4(12 - x) = 28$

2x + 48 - 4x = 28

$2x = 20[M1]$

$x = 10, y = 2$ **[A1]**

[Total Marks: 7]

14 (a) (i) 34 **[B1]**

(ii) $48 - 26$ **[M1]**

22 **[A1]**

(b) (i) 0.25×240 **[M1]**

60 **[A1]**

(ii) For Company X, 22 and 38 are the lower and upper quartiles, so 50% of 160 = 80 deliveries took between 22 and 38 minutes **[B1]**

For Company Y, 22 and 38 are not quartiles, so the exact proportion of deliveries in this interval cannot be determined from the box plot **[B1]**

Since Company Y had more total deliveries (240 > 160), Company Y could have more than 80 deliveries in that interval **[B1]**

(c) Company X **[B1]**

Company X has a lower median delivery time (30 minutes < 34 minutes), so it is faster on average **[B1]**

Company X has a smaller interquartile range (16 minutes < 22 minutes), so its delivery times are more consistent **[B1]**

[Total Marks: 11]

15 **(a)** Upper Quartile = 44 and Lower Quartile = 26 **[M1]**

$44 - 26 = 18$ **[A1]**

(b) The Adults generally completed the puzzle faster because their median time is lower ($32 < 36$) **[B1]**

The Students' times were more consistent (less spread out) because their interquartile range (12) is smaller than the Adults' interquartile range (18) **[B1]**

(c) The reasoning is invalid because range only measures spread, not the actual values or maximum times **[B1]**

However, the conclusion happens to be true because the maximum time for Students (48) is less than the maximum time for Adults (55) **[B1]**

[Total Marks: 6]

16 **(a)** Forms a correct equation: $x + 2x + 120 + x = 360$ **[M1]**

Solves to find $4x = 240$

x = 60$[M1]$

Calculates number of students: $\frac{60}{360} \times 120 = 20$ **[A1]**

(b) Calculates the percentage for Class P: $\frac{120^\circ}{360^\circ} \times 100 = 33.3\%$ (or $\frac{1}{3}$) **[M1]**

Identifies total ratio parts for Class Q $= 1 + 3 + 2 + 2 = 8$ **[M1]**

Calculates percentage for Class Q: $\frac{3}{8} \times 100 = 37.5\%$ and concludes $37.5\% > 33.3\%$ **[A1]**

(c) The two classes have different total numbers of students (so proportions/percentages must be used for a fair comparison). **[B1]**

[Total Marks: 7]

17 **(a)** 112 **[B1]**

(b) 120 and 104 seen as quartiles **[M1]**

16 **[A1]**

(c) Cumulative frequency of 110 seen, leading to $120 - 110 = 10$ apples **[M1]**

8.33 (8.333...) **[A1]**

(d) Farm X apples are heavier on average as their median (112 g) is greater than Farm Y (105 g). **[B1]**

Farm X apples are more consistent in mass / less spread out as their interquartile range (16 g) is smaller than Farm Y (22 g). **[B1]**

[Total Marks: 7]

18 **(a)** **(i)** 136 to 138 **[B1]**

(ii) Upper quartile (90th) = 146 to 148 AND Lower quartile (30th) = 126 to 128 **[M1]**

18 to 22 **[A1]**

(b) Reads cumulative frequency at 150g as 98 (allow 97 to 99) **[M1]**

$120 - 98 = 22$ (allow 21 to 23) **[A1]**

(c) Number of apples $\leq$ 150g is 98 (or their read value from (b)) **[B1]**

$\frac{98}{120} \times \frac{97}{119}$ **[M1]**

$\frac{9506}{14280}$ or $\frac{4753}{7140}$ or 0.666 $(0.6656\ldots)$ **[A1]**

[Total Marks: 8]